

# Micro- and Nanorobots for Targeted Drug Delivery in Childhood Cancer: Emerging Technologies, Clinical-Trial Evidence, and Translational Priorities

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## Abstract

**Background:** Micro- and nanorobotic delivery platforms aim to combine controlled movement or molecularly triggered actuation with localized therapeutic delivery. Their potential in childhood cancer must be distinguished from the more established field of nonrobotic nanomedicine. Pediatric nanocarriers have reached clinical investigation, but this does not establish the clinical readiness of actively controlled robots.[1]

**Objective:** To examine the state of micro- and nanorobotic drug delivery relevant to childhood cancers, distinguish pediatric evidence from broader oncology demonstrations, and identify requirements for clinical translation.

**Methods:** Targeted academic searches were supplemented with web and ClinicalTrials.gov searches. Selected primary studies and technology reviews were evaluated for platform design, payload, experimental setting, clinical maturity, and translational limitations. Full text was used when available; several key pediatric reports could only be assessed from abstracts. No pooled effect estimate was calculated.

**Results:** Direct pediatric-relevant evidence included a magnetic diatom microrobot platform for neuroblastoma with in-vitro therapeutic evaluation and a microrobotic catheter platform for retinoblastoma validated in patient-derived vascular phantoms.[2, 3] Broader oncology evidence included molecularly triggered DNA-origami robots producing tumor-vessel thrombosis in mice, while recent magnetic delivery systems demonstrated imaging-compatible navigation in large animals.[4, 5] No interventional pediatric cancer trial of an actively navigated micro/nanorobot or molecularly actuated DNA nanorobot was identified in the retrieved evidence. A pediatric rhenium-186 nanoliposome trial illustrates adjacent clinical nanomedicine, not robotic delivery.[6]

**Conclusion:** Micro- and nanorobotic delivery is an emerging preclinical approach, not an established treatment for childhood cancer. The most defensible development pathway requires pediatric-relevant efficacy models, imaging and navigation validation, biodistribution and clearance studies, reproducible manufacture, and explicit evidence that robotic actuation adds value beyond the same payload delivered without actuation.[5, 7]

**Keywords:** pediatric oncology; childhood cancer; microrobots; nanorobots; targeted drug delivery; neuroblastoma; retinoblastoma; DNA origami; magnetic actuation.

## 1. Introduction

The central question is not whether very small carriers can transport anticancer drugs, but whether robotic functions improve delivery enough to justify their added biological and engineering complexity. Pediatric nanodelivery research already encompasses targeted formulations and clinical investigations, including applications to neuroblastoma, sarcomas, and brain tumors.[1] Micro- and nanorobotic systems add functions such as magnetic navigation, biologically generated propulsion, or molecularly triggered structural opening. Contemporary reviews describe these mechanisms as promising while emphasizing unresolved clinical translation problems.[7, 8]

This review asks three linked questions: Which platforms have direct relevance to childhood cancer? What has actually been demonstrated in human trials? Which engineering advances could support a credible pediatric development program? The distinction between a technology demonstration and a clinical treatment is maintained throughout.

## 2. Methods and Scope

Academic discovery used complementary queries covering recent cancer micro/nanorobotics, childhood-cancer applications, and DNA-, bacterial-, and magnetically mediated delivery. Web discovery and a ClinicalTrials.gov search

supplemented the literature search. Sources were prioritized for direct pediatric relevance, primary experimental evidence, clinical navigation demonstrations, or clarification of translational barriers. Selected full texts were examined using structured extraction of propulsion, payload, experimental model, and limitations.

The synthesis is a targeted narrative review. It does not use a registered protocol, formal risk-of-bias instrument, exhaustive registry screening, or meta-analysis. Registry identification relied on terminology associated with micro-robots and nanorobots; systems registered under other device or product names could therefore have been missed. The neuroblastoma diatom, retinoblastoma catheter, and foundational DNA nanorobot reports were assessed from available abstracts because full texts were not recovered. This constrains evaluation of dosing, controls, reproducibility, and detailed toxicity results. Consequently, statements about absent pediatric trials mean “not identified in this search,” not definitive proof that none exist worldwide.

### 3. What Qualifies as Robotic Drug Delivery?

For this review, a robotic platform must have controlled actuation: externally guided movement, self-propulsion with a targeting or navigation function, or programmed mechanical reconfiguration that controls payload exposure. This operational definition includes molecularly triggered DNA-origami structures even when they do not swim. The foundational DNA example uses an aptamer to trigger mechanical opening and expose thrombin, illustrating why propulsion alone is too narrow a definition.[4]

A conventional liposome, nanoparticle, or convection-enhanced infusion is not classified here as a robot merely because it is small or locally delivered. For example, the pediatric rhenium-186 study uses a liposomal formulation infused through catheters; its registry description does not describe autonomous movement or robotic actuation.[6] Conversely, a robotically controlled microcatheter is a relevant delivery-interface technology, but it is not a free-swimming nanorobot. These distinctions prevent ordinary nanomedicine trials from being misrepresented as evidence for robotic cancer treatment.

### 4. Direct Evidence Relevant to Childhood Cancers

#### 4.1 Neuroblastoma: magnetic diatom microrobots

A recent neuroblastoma study constructed magnetic microrobots from diatom frustules, using their hierarchical porous structure as a delivery scaffold. Amino-group and folic-acid functionalization supported drug conjugation and targeting. The reported platform carried cisplatin or paclitaxel formulations with pH-responsive release, and magnetic testing demonstrated controlled motion.[2]

The abstract reports enhanced therapeutic efficacy and reduced off-target effects in vitro.[2] These results constitute pediatric-disease-specific proof of concept, but not evidence of improved survival, reduced organ toxicity, or benefit in treated children. The accessible abstract does not provide sufficient detail to reconstruct the comparator design or establish whether motion itself, targeting chemistry, altered release, or some combination explains the reported improvement.

The appropriate next question is therefore incremental benefit: does magnetic actuation improve tumor exposure and therapeutic activity relative to an otherwise identical stationary carrier? A pediatric development program should not assume that a favorable cell-culture result demonstrates navigability in blood or penetration into a solid tumor.

#### 4.2 Retinoblastoma: microrobotic catheter navigation

A recent retinoblastoma-related report describes teleoperated and autonomous navigation of flow-driven micro-catheters into the ophthalmic artery. Its platform combines real-time imaging, closed-loop control, and miniaturized instrumentation, with navigation strategies informed by computational models of pediatric vascular anatomy.[3]

The reported validation used patient-derived biomimetic phantoms on the benchtop.[3] This is a clinically motivated engineering demonstration, not a pediatric treatment trial. It also occupies a different technological category from a drug-loaded mobile particle: the robot controls the delivery route rather than acting as the therapeutic carrier itself.

This distinction matters when assessing readiness. A catheter-navigation system can potentially build on a localized delivery procedure, whereas a freely circulating carrier must additionally establish material fate, distribution, and

elimination. Whether either approach improves patient outcomes requires clinical evidence, not navigation accuracy alone.

## **5. Broader State-of-the-Art Technologies and Their Pediatric Applicability**

### **5.1 Imaging-compatible magnetic delivery**

A recent magnetic microrobot platform combines a gelatin matrix, magnetic zinc-doped iron oxide particles, radiopaque tantalum, and a therapeutic agent. The design integrates magnetic control with visibility under clinically available imaging.[5] Experimental payload work included doxorubicin and ciprofloxacin, while a separate vascular-model demonstration delivered a thrombolytic payload to a clot.[5]

Large-animal experiments demonstrated fluoroscopic tracking in cranial arteries and magnetic navigation in a cerebrospinal-fluid setting.[5] These results are important because they test integrated navigation beyond a simple culture dish. Nevertheless, they do not establish treatment efficacy in pediatric cancer. The paper itself identifies long-term safety, scalability, and automated navigation as priorities for further development.[5]

The translational lesson is architectural: drug carrier, release device, guidance system, and imaging must function together. “Clinically ready” in a publication title should not be read as equivalent to approved clinical use or demonstrated benefit in children.

### **5.2 Molecularly triggered DNA nanorobots**

DNA-origami robots offer a different form of control. In a foundational cancer study, a nucleolin-binding aptamer served as both targeting ligand and opening trigger. The robot carried thrombin internally and exposed it following activation, inducing coagulation in tumor-associated vessels.[4]

Tumor-bearing mouse experiments demonstrated vascular thrombosis, tumor necrosis, and inhibition of tumor growth. The investigators also reported favorable safety and immunological findings in mice and miniature pigs.[4] These observations support a mechanism in the studied models, not a conclusion that the strategy is safe for childhood cancers.

For pediatric translation, the proposed benefit and hazard are closely linked: local coagulation is the therapeutic mechanism, so inappropriate activation must be treated as a central safety concern. A future program would need to verify target accessibility in the intended pediatric tumor and rigorously distinguish tumor-restricted activation from systemic exposure. These are development requirements proposed here, not findings of a completed pediatric trial.

### **5.3 Biohybrid, acoustic, and light-responsive systems**

Current reviews describe bacterial or cell-based carriers, ultrasound-driven motors, and light-responsive propulsion as additional preclinical approaches. Such platforms have delivered drugs, proteins, or nucleic-acid payloads in experimental models, but results from human-derived cells should not be confused with treatment of human participants.[7, 9]

Their pediatric applicability remains conditional. Biohybrid carriers require careful evaluation of immune interactions and payload toxicity. Chemically powered systems must address fuel and propulsion-byproduct compatibility, while optical systems are constrained by tissue penetration. Real-time tracking also remains an important translational requirement.[7, 9]

No single propulsion mechanism is therefore universally “best.” The preferred mechanism must be evaluated against the intended route, anatomical depth, required payload, imaging method, and acceptable material persistence. This is an engineering selection framework rather than a ranking established by comparative pediatric trials.

## **6. Clinical-Trial Evidence: What Has Reached Children?**

No interventional pediatric cancer study of the robotic platforms defined in Section 3 was identified in the retrieved records. The directly relevant neuroblastoma report describes in-vitro work, and the retinoblastoma report describes

benchtop phantom validation.[2, 3] More generally, recent technology reviews emphasize the remaining gap between laboratory demonstrations and actual patient treatment.[7]

An adjacent example is NCT07061626, a pediatric study of rhenium-186 obisbameda, described in the official title as a two-part phase 1/2a investigation in recurrent, refractory, or progressive ependymoma and high-grade glioma. The retrieved registry record specifies participants aged 6–21 years and evaluation of dose tolerance, response, and progression-free survival.[6] The formulation delivers encapsulated rhenium locally by convection-enhanced delivery.[6]

This trial is relevant as an example of clinical pediatric nanodelivery, but it is not evidence for nanorobotic treatment. The retrieved record labels the study “not yet recruiting”; this is the status of the available registry record, not independently confirmed current recruitment information.[6] Planned enrollment and dates cannot be interpreted as treated patients or clinical results.

The key evidence gap is thus specific: pediatric studies demonstrating added therapeutic benefit from robotic actuation, with acceptable distribution, clearance, and safety, were not identified. Broad statements that “nanorobots are already treating childhood cancer in clinical trials” are not supported by this evidence set.

## 7. Translational Barriers

### 7.1 Navigation and targeting are different endpoints

A carrier can move reproducibly without delivering sufficient drug to viable tumor tissue. Reviews explicitly note that self-propulsion does not guarantee selective targeting and emphasize integration with imaging.[9] Evaluation should therefore separate navigation success, regional deposition, tumor uptake, intracellular delivery, and anti-cancer activity. These are proposed assessment categories; none should be used as a substitute for the next without validation.

### 7.2 Material clearance and long-term safety

Magnetic materials must balance actuation performance against biocompatibility and biodegradability. The magnetic platform report identifies this trade-off and calls for long-term safety studies.[5] Demonstration of acceptable cell viability or short-term animal observations does not establish chronic safety in children.

For a pediatric program, proposed safety work should include drug and carrier biodistribution, persistence after payload release, immune and coagulation effects, tissue injury associated with the navigation procedure, and developmental outcomes relevant to the intended population. The emphasis should depend on mechanism: a thrombin carrier and a magnetic catheter system require different hazard models.

### 7.3 Manufacturing and integrated-system reliability

A robotic medicine is an integrated product, not merely a drug formulation. Release reproducibility, magnetic or molecular responsiveness, imaging visibility, sterility, and mechanical reliability all require specification. Current literature identifies scalability and quality control as unresolved development priorities.[5, 7] Using previously approved constituent materials does not establish approval of the assembled robot or its drug-device combination.

## 8. Proposed Research and Clinical Development Framework

The following is a proposed roadmap, not a trial that has been conducted or registered.

**First, establish disease-specific biological benefit.** Use models that match the intended pediatric indication and compare the same therapeutic payload across free drug, stationary carrier, actuated carrier, and an empty-carrier control where feasible. The main question should be whether actuation adds meaningful benefit, not simply whether the complete platform can kill cancer cells.

**Second, demonstrate deliverability under clinically relevant conditions.** Test navigation in realistic geometry and flow, then connect localization to quantitative tumor exposure and release. Imaging compatibility should be

demonstrated with the actual clinical guidance system rather than inferred from microscopic tracking. This recommendation follows the literature's emphasis on clinical visibility and tracking.[5, 7]

**Third, establish an integrated safety package.** Evaluate carrier fate separately from drug pharmacology and investigate the failure modes specific to the chosen mechanism. Long-term safety and scalable manufacture should be development gates rather than deferred assumptions.[5]

**Fourth, design an early clinical study around feasibility and safety.** If preclinical requirements are satisfied, a proposed early-phase study should prespecify navigation success, dose-delivery verification, acute adverse events, carrier clearance, and procedure-related injury. Tumor response would be exploratory initially. Longitudinal follow-up should evaluate delayed effects appropriate to the drug, material, anatomy, and age group.

A sample size, dosing schedule, or enrollment age range cannot be responsibly specified without selecting a platform, indication, route, and supporting toxicology package. Adult or large-animal feasibility may inform development, but should not automatically be treated as a mandatory or sufficient bridge for every pediatric indication.

## 9. Discussion

The most important recent advance is integration: movement, release, and tracking are increasingly evaluated within a single delivery system, including large-animal navigation work.[5] Direct childhood-cancer relevance is also emerging through neuroblastoma carriers and retinoblastoma catheter-navigation prototypes.[2, 3] However, these advances address different problems and cannot be combined into one claim of clinical efficacy.

The evidence supports a distinction between pediatric nanomedicine, pediatric microrobotic proof of concept, and nonpediatric preclinical robotic delivery. Nanoliposomal clinical investigations demonstrate that nanoscale formulations can enter pediatric development; they do not validate propulsion, robotic targeting, or autonomous molecular operation.[6]

A credible future program should make a falsifiable claim: adding robotic actuation improves tumor delivery or therapeutic benefit compared with a matched nonactuated system, without unacceptable new hazards. Without that comparison, improved performance may arise from the carrier chemistry or release mechanism rather than robotics itself.

## 10. Conclusion

Micro- and nanorobots provide experimentally demonstrated approaches to controlled delivery, including magnetic guidance and molecularly triggered payload exposure. In childhood-cancer-relevant research, current retrieved evidence consists of in-vitro neuroblastoma delivery and benchtop retinoblastoma navigation, not demonstrated clinical benefit in children.[2, 3] Broader DNA and magnetic platforms strengthen the technological rationale but remain insufficient evidence for pediatric efficacy or long-term safety.[4, 5]

The defensible conclusion is that robotic drug delivery for childhood cancer is a promising research direction with a substantial translational gap. Its next milestones should be matched-control efficacy, clinically compatible imaging and navigation, carrier-clearance validation, and reproducible manufacture—not claims of readiness based solely on particle size or motion.

## Figures and Research Hypothesis

Figure 1. Evidence setting of selected delivery platforms

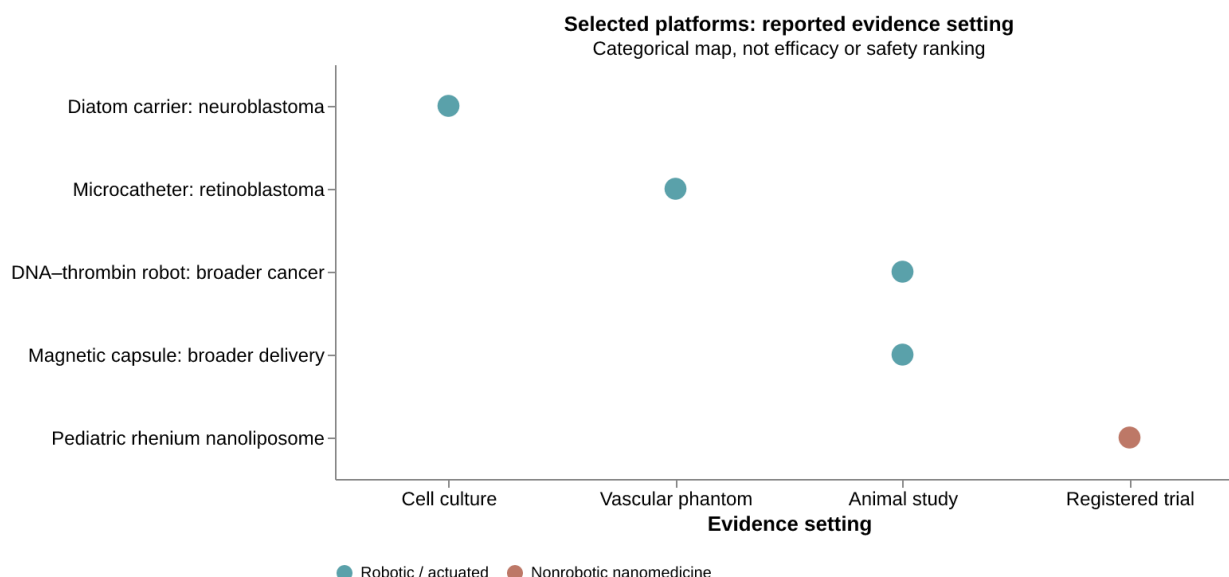


Figure 1. Stages describe the highest setting identified in the selected reports, not numerical scores or a complete field-wide pipeline. A registered trial does not imply completed treatment or positive results. Pediatric microrobotic examples remain in cell or phantom settings.

The neuroblastoma platform was assessed in vitro, whereas the retinoblastoma system was validated in vascular phantoms. DNA and magnetic platforms have animal evidence, but the registered pediatric nanoliposome trial uses nonrobotic delivery. The stages are categorical and do not quantify comparative efficacy or safety.[2–6]

Figure 2. Selected planned pediatric trial cohorts

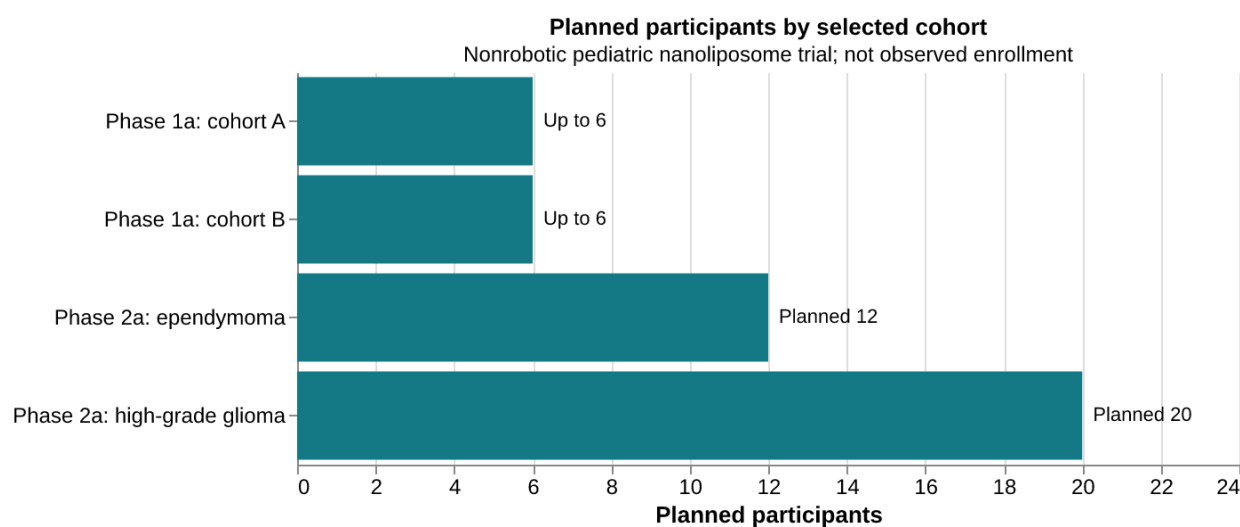


Figure 2. Selected protocol cohorts in NCT07061626, not actual enrollment or outcomes. Phase 1b is omitted; these bars are not the full trial total. This is nonrobotic convection-enhanced delivery.

The protocol specifies up to six participants in each initial cohort, followed by planned expansion groups of 12 participants with ependymoma and 20 with high-grade glioma. These are planned numbers, not actual enrollment.

Phase 1b is not shown, so the displayed cohorts must not be summed as the complete trial population. This trial concerns nonrobotic nanoliposomes.[6]

Figure 3. Protocol radiation-dose estimates

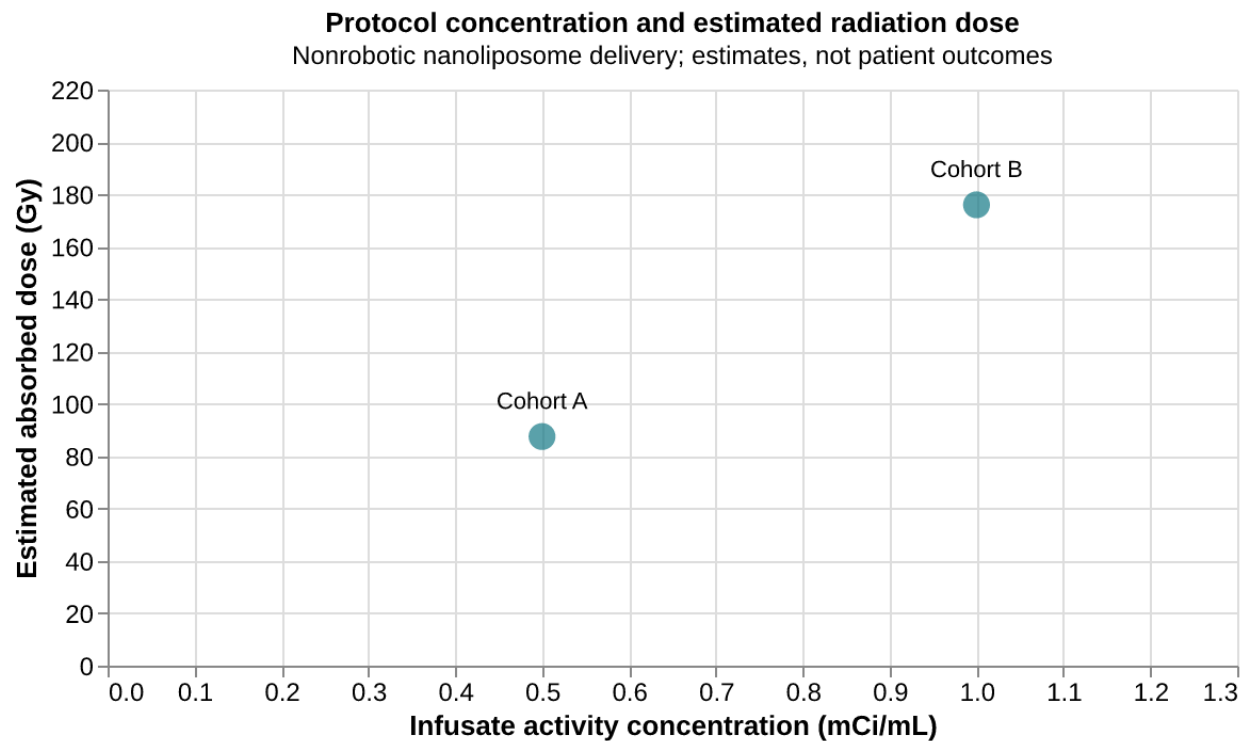


Figure 3. *Protocol-specified infusate concentrations and approximate absorbed-dose estimates for two pediatric nanoliposome cohorts. Not measured patient doses, efficacy results, or a validated concentration–dose curve; no line is fitted. Nonrobotic delivery.*

For initial cohorts A and B, the protocol pairs infusate activity concentrations of 0.5 and 1.0 mCi/mL with approximate absorbed-dose estimates of 87.5 and 176 Gy, respectively. These are protocol estimates for local radionuclide delivery—not measured patient doses, chemotherapy doses, or observed benefit. No concentration–dose curve or statistical uncertainty has been inferred from these two points.[6]

### Summary of the hypothesis

**Proposed primary hypothesis:** In pediatric-relevant neuroblastoma models, magnetically actuated drug-loaded microrobots increase tumor drug exposure and antitumor activity compared with the same carrier and payload without actuation, without increasing normal-tissue injury.

This is a testable research proposition, not an established clinical result. Its rationale is the reported in-vitro neuroblastoma activity of magnetic diatom carriers and the broader demonstration that integrated magnetic systems can be tracked and navigated in animal settings. Those studies do not establish that actuation itself produces superior outcomes in children.[2, 5]

**Null hypothesis:** Actuation provides no improvement in tumor drug exposure or antitumor activity over the matched nonactuated carrier; alternatively, any delivery gain is offset by additional normal-tissue injury.

**Proposed test:** Compare free drug, a stationary loaded carrier, the matched actuated loaded carrier, and empty-carrier controls. Prespecify tumor drug exposure as the delivery endpoint, tumor growth or viable tumor fraction as efficacy endpoints, and normal-tissue injury and carrier persistence as safety endpoints. Keep payload dose and

carrier composition matched between stationary and actuated groups. Outcomes should be reported separately rather than combined into an arbitrary score.

**Interpretation rule:** Higher tumor exposure alone would support improved delivery, not clinical benefit. Greater antitumor activity without unacceptable added injury would support further preclinical development. A negative matched-control result would weaken the claim that robotics adds value. Persistent material, off-target deposition, or procedure-related harm would require redesign even if tumor uptake improved. Long-term safety and clearance remain important unresolved requirements.[5]

**Data limitation:** The retrieved evidence did not supply comparative pediatric patient response, survival, or adverse-event data for robotic delivery. Such curves cannot be constructed responsibly from trial plans or cell-culture findings. Accordingly, the quantitative clinical plots show nonrobotic protocol data and the robotic chart shows evidence settings, not invented patient outcomes.

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